

Build a review-ready UCC microcredential

A six-stage tutorial with completion indicators, field-by-field examples and a complete illustrative course.

A worked model—not a prescribed course

Every field is filled to show the expected level of clarity. Facilitators should adapt the evidence, content, dates and assessment to their own discipline and obtain academic approval before delivery.

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What this guide gives you

- A clean route through all six Course Studio stages.
- Plain-language explanations of required fields and optional enhancements.
- A complete model course that can be loaded safely into the unsaved editor.
- Checks for outcome alignment, accessibility, authentic evidence and the certificate gate.
- A final pre-submission checklist for academic approval.

Start in the application

Open Facilitator Portal → Course Studio → Load complete example. The example changes only the current unsaved editor state until you choose Save draft.

One course, six visible stages

The redesigned workspace separates planning, content, practice, assessment and final review. The right-side checks show what is complete and what must be corrected before the draft can be submitted.

Stage	Purpose	Completion evidence
1 · Essentials	Define the offer and audience.	Code, title, description, discipline, level, language, delivery, hours, enrolment and price.
2 · Outcomes	Connect purpose, capability and evidence.	Objectives, measurable outcomes, skill, assessment method, skills tags and course sections.
3 · Learning	Create the learner journey.	Accessible learning blocks mapped to a section and at least one outcome.
4 · Activities	Add authentic practice.	Colab or virtual-lab instructions, rubric, pass mark, attempts, due date and required/optional status.
5 · Assessment	Configure valid judgement.	Assessment modes, scored questions, feedback, marking schemes, pass mark and attempts.
6 · Review	Test and submit with confidence.	Readiness checks, learner preview, saved draft and submission for academic approval.

Read the indicators

Indicator	Meaning	Action
Green tick	Every required check for the stage is satisfied.	Continue, but still review quality and accuracy.
Clock / incomplete	A named required field or evidence item is missing.	Open the stage and follow the exact side-check message.
Optional	The enhancement may be omitted.	If added, configure it fully and ensure it serves an outcome.
Readiness score	A design-quality signal, not academic approval.	Resolve all blocking checks before submission.

Side-check rule

A completed field is not the same as a good field. Use specific language, observable verbs and evidence that a reviewer can inspect.

Define the offer, then align the design

Stage 1 • Essentials

Field	Illustrative entry	Why it works
Course title	Applied Data Literacy for Evidence-Based Decision-Making	Names the capability and practical purpose.
Code	UCC-MC-ADL-001	Unique, short and institutionally traceable.
Audience	Public-sector, NGO, academic and business professionals who interpret data.	Identifies who will use the learning and in what context.
Prerequisites	Basic computer and spreadsheet skills; no programming experience required.	Sets fair expectations without hidden barriers.
Duration / mode	30 hours · blended · applied level	States learner workload and delivery pattern.
Enrolment / price	Application · GHS 450	Makes access and commercial conditions explicit.
Accessibility	Structured HTML, heading-led documents, reviewed transcripts and keyboard support.	Describes concrete provision, not a generic promise.

Stage 2 • Outcome-led design

Write objectives as broad intentions. Write outcomes as observable learner performances and name the evidence that will prove achievement.

Outcome statement	Skill	Assessment method
Explain the data lifecycle and identify quality, provenance, privacy and ethical risks.	Conceptual understanding	Objective knowledge check
Clean, summarise and visualise a small dataset in Google Colab using a reproducible workflow.	Data literacy	Applied practical evidence
Interpret a chart accurately and communicate a finding with uncertainty and limitations.	Professional communication	Presentation / demonstration
Defend an ethical, evidence-informed recommendation for a realistic decision.	Ethical decision-making	Oral assessment

Build an accessible learner journey

The example uses four sections and seven learning blocks. Every block belongs to a section, has a clear learner-facing title, names its source and maps to one or more outcomes.

Section	Learning block	Format / design note
1 · Orientation	Start here: pathway, support and certificate conditions	Readable HTML · states pathway and credential rules.
1 · Orientation	Annotated community-indicators case pack	PDF download · includes data dictionary and decision prompts.
2 · Foundations	Data quality, provenance and ethical use	Readable HTML · five pre-analysis checks.
2 · Foundations	How a neural network learns	Reviewed external video · link plus companion transcript.
3 · Applied analysis	Worked example: community-service dataset	Readable HTML · question-to-chart workflow.
3 · Applied analysis	World Bank Open Data explorer	External embed/link · source terms checked before publication.
4 · Capstone	Capstone evidence and oral-defence brief	Editable Word plus equivalent readable HTML.

Choose the right content method

Use	When	Minimum check
Embed readable text	The content is course-authored or can be lawfully adapted.	Headings, lists, links and plain-language structure render correctly.
Upload PDF / Word	Learners need a fixed or editable reference file.	Retain a readable HTML equivalent; never display raw binary text.
Link / embed	A stable authoritative external source adds value.	Open in a new tab, name the source and provide an alternative if embedding fails.
Video	Motion or demonstration serves an outcome.	Review captions/transcript, duration, relevance and access conditions.

Document preview safeguard

The learner preview must render extracted readable text or offer the original download. PDF or Word bytes must never be decoded directly into the lesson window.

Require authentic practice and valid evidence

Stage 4 • Programme activities

Activity	Gate	Configuration
Colab data-cleaning and visualisation notebook	Required	Pass 70% · 3 attempts · instructions, notebook and 100-mark rubric supplied.
Virtual measurement and data-quality practical	Required	Pass 70% · 2 attempts · safety gate, observations, calculation and report rubric.
Ohm's law data interpretation	Optional	Pass 60% · 3 attempts · enrichment only; not part of the certificate gate.

Colab preview: two supported routes

1. A valid GitHub .ipynb or Google Colab URL opens directly in a new tab.
2. An uploaded or bundled notebook opens Colab and downloads the .ipynb file; in Colab choose File → Upload notebook.
3. If nothing opens, allow pop-ups for the UCC platform and try Open activity again.

Stage 5 • Assessment

The example demonstrates the platform's complete question repertoire. Use only formats that provide valid evidence for the mapped outcome.

Assessment modes (8)	Question formats demonstrated (13)
Objective quiz; applied assignment; practical evidence; presentation; demonstration; portfolio; oral assessment; reflective evidence	Multiple choice; true/false; fill in; matching; drag and drop; picture matching; video question; short answer; essay; scenario response; oral-defence prompt; evidence-upload prompt; practical assignment

- Every scored item includes points, a marking scheme, correct/incorrect feedback and learner advice.
- Every question maps to at least one outcome.
- The course pass mark is 70%; three attempts are allowed.
- Required practical evidence is checked separately from the objective assessment score.

Review, preview and submit

Use Review as a quality-assurance workspace. It should show the learner sequence, outcome mappings, assessment evidence and the credential gate in one place.

Review check	Evidence of completion
Identity and catalogue	Course code and title are final; audience, prerequisites, price and workload are accurate.
Alignment	Every outcome has a skill and assessment method; learning and assessment evidence map back to outcomes.
Learner journey	Sections are ordered; titles are meaningful; duration is realistic; links and downloads open.
Accessibility	HTML alternatives, headings, transcripts, labels and keyboard operation are checked.
Activities	Instructions, rubric, pass mark, attempts, due date and gate status are complete.
Assessment	Questions, solutions, feedback and marking schemes have been tested in learner preview.
Credential	Certificate is enabled only when the course has an approved, defensible completion rule.
Governance	Save draft, preview as student, then submit for academic review; publication remains an administrator action.

Certificate gate

Verified identity + at least 70% on the course assessment + completion of every required programme activity = automatic University of Cape Coast digital certificate with a live verification QR.

Before selecting Submit for review

- Replace all illustrative titles, codes, dates, people, datasets and links.
- Confirm copyright, licence, privacy and consent for every external or uploaded resource.
- Preview every lesson and activity as a student on desktop and mobile.
- Ask another facilitator to complete the instructions without verbal explanation.
- Resolve every blocking side check and record any justified accessibility alternative.

Applied Data Literacy for Evidence-Based Decision-Making

This model is intentionally complete so facilitators can study the relationship between purpose, curriculum, authentic practice, assessment and credentialing.

Design element	Illustrative specification
Discipline / level	Professional · Applied
Delivery / language	Blended · English
Workload / access	30 hours · application enrolment · GHS 450
Objectives / outcomes / skills	4 objectives · 4 measurable outcomes · 8 skills tags
Curriculum	4 sections · 7 learning blocks
Practice	3 activities: two required and one optional
Assessment	8 modes · 13 question types · 70% pass mark · 3 attempts
Credential	UCC digital certificate with QR verification after all gate conditions are met

Four course objectives

1. Build practical confidence in asking answerable questions and judging data quality, provenance and limitations.
2. Develop the ability to clean, summarise and visualise a small dataset with transparent, reproducible methods.
3. Enable ethical communication of evidence-based recommendations to academic, workplace and community audiences.
4. Strengthen authentic assessment through process evidence, reflection and a short oral defence.

Eight skill tags

Critical thinking • Digital literacy • Evidence-based decision-making • Applied problem-solving • Communication • Data literacy • Research and inquiry • Responsible data storytelling

Use the assets

The ZIP includes a PDF case pack, editable Word capstone brief, Colab notebook and CSV question bank. They are examples for study and must be adapted before a live course is published.

Course submission checklist

Done	Quality assurance statement
☐	The title, code and description clearly distinguish this course.
☐	Audience, prerequisites, workload, delivery, enrolment and price are accurate.
☐	Objectives describe purpose; outcomes use observable verbs and name evidence.
☐	Every learning block belongs to a section and maps to an outcome.
☐	PDF/Word items have a readable alternative and external links have been tested.
☐	Videos have reviewed captions or a transcript and all images are meaningfully labelled.
☐	Each programme activity has instructions, rubric, pass mark, attempts and gate status.
☐	Questions provide enough evidence and include marking guidance and feedback.
☐	The certificate gate matches the approved achievement standard.
☐	Student preview works on desktop/mobile, including downloads, Colab and virtual labs.
☐	All illustrative data and placeholder dates have been replaced.
☐	A colleague has reviewed the full learner journey before submission.

Files supplied with this tutorial

- UCC-Facilitator-Course-Studio-Annotated-Guide.pdf and .docx
- Applied-Data-Literacy-Case-Pack.pdf
- Applied-Data-Literacy-Capstone-Brief.docx
- Applied-Data-Literacy-Colab-Notebook.ipynb
- Applied-Data-Literacy-Question-Bank.csv

Institutional note

The platform can generate a credential only after the configured evidence is complete. Academic approval and the authority to issue a University of Cape Coast credential remain governed by University policy.